Aviaee - Air-based Delivery System

Team Members: Sahiti Akella, Osama Al Riyami, Hooriya Amjad, Teresa Lavoie, Vera Sokolosky
Academic Advisor: Norm Bartley
Sponsoring Company: Aviaee

Motivations

- A fundamental component of our society is the timely delivery of materials, components, and packages over land
- As demand increases, it becomes more challenging for service providers to maintain standards at current prices
    - Investment into roads is costly and time-consuming
    - Routing is not always flexible due to road constraints
    - Environmental impacts

Problem statement: The current model of ground transportation is unsustainable [1] as populations and demand increase.

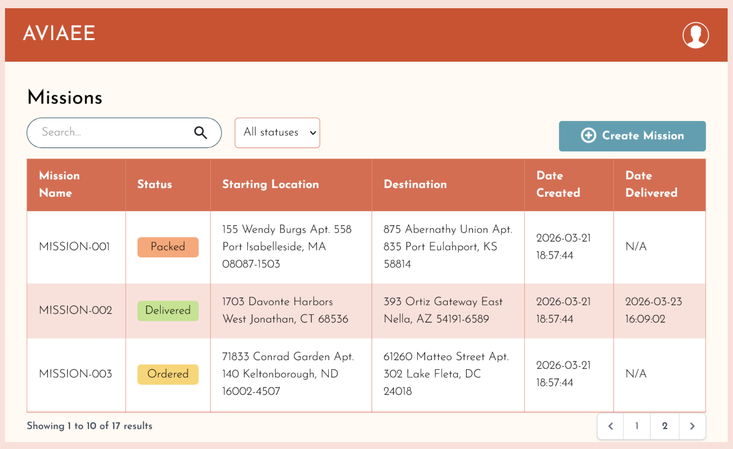
Solution

- Aviaee Inc. is developing a software tool and drone that allows an automatic scheduling system to activate unmanned aerial vehicles beyond the visual line of sight in the air domain. This allows for:
    - Minimized transportation time
    - Increased accessibility to remote and rural locations
    - Streamlined resource requirements
    - Limited reliance on expensive infrastructure
    - Reduction in vehicle costs

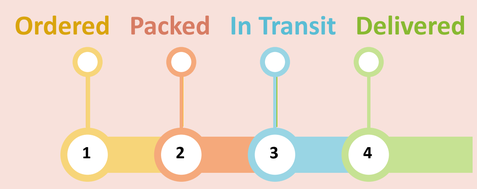
- We have contributed to the development of an Air Based Delivery System, which consists of software and hardware components:
    - A web-based platform to host and track “Mission” data
        - **Mission:** The drone delivery trip consisting of the pickup of the package from the seller, and the drop-off to the buyer
    - A grabbing tool prototype
    - Mission statuses are communicated within the software, and are referenced by users (pilots) to determine when the drones and their grabbing tools are ready to be operated

Web Platform

One of the two major components of this project is a web platform and corresponding database with tables of missions and users.



The web portal allows users to register their account as one of three user types: Buyers, Sellers, or Pilots. Once successfully logged in, the user is taken to the Missions page where they can view, create, update, and delete missions.



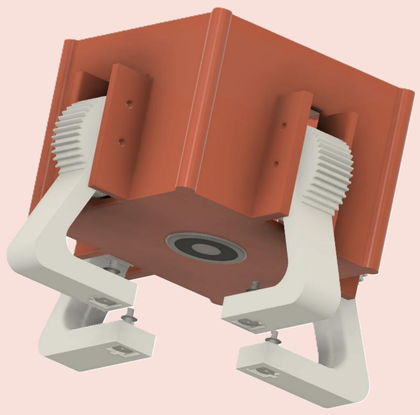
- Web platform implements the clean architecture with Laravel
    - This architecture makes the code more expandable and easy to maintain in the future
- Database stores mission and user data
- User interface and experience was designed with intent to display data in a way that is concise, informative, and engaging
    - Implemented validation and exception handling to ensure meaningful user and developer feedback

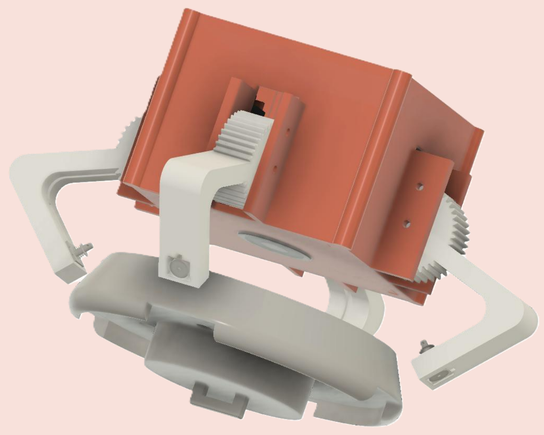
Grabbing Tool

The other major component of this project is a 4-arm grabbing tool prototype to be added to the company’s existing drone in the future.

The grabbing tool was designed to be able to close its arms around a specialized package attachment and securely hold it during transit.

- Design incorporates a magnet and solenoid pins
    - The magnet helps align the grabbing tool to the attachment during package pickup, and helps keep the attachment attached to the grabbing tool during flight
    - Solenoid pins were added to improve the hold on the package in case of power loss
- Circuit design integrated with existing drone’s power and control system
    - Power is taken from the drone battery
    - Grabbing tool consumes minimal power and has a negligible impact on flight time





Next Steps

After concluding our capstone project, we considered how the software and hardware will connect in the future, and other/extra deliverables that we would like to implement from a design and engineering perspective:

- Connecting the grabbing tool/drone to the web platform
    - Metadata from drone will come into web platform
        - Position of the drone relative to mission statuses
        - Video/image data transferred to web platform
- Add additional features to the grabbing tool for improved functionality
    - Weight sensor and distance sensor to be added to the grabbing tool for improved functionality and redundancy when picking up packages
- Establish user roles and privileges to distinguish permissions between buyers, sellers, and pilots
    - Only buyers will be able to create missions
    - Sellers will be associated with a shop address
    - Pilots and sellers can edit mission statuses
- Payment portal for users to purchase their delivery
    - Once user roles are implemented, buyers will be able to use a payment portal to purchase missions
    - The payment portal will be used to track and manage these payments

References

[1] US EPA,OAR, “Why Freight Matters to Supply Chain Sustainability | US EPA,” US EPA, May 26, 2016. https://www.epa.gov/smartway/why-freight-matters-supply-chain-sustainability